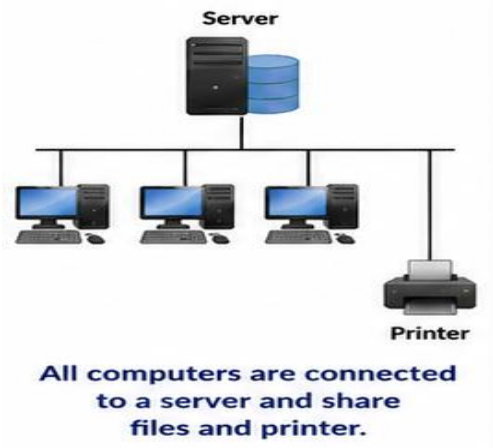
What is a Computer Network?

A computer network is a group of two or more computers and other devices connected through communication links or cables to exchange data and share resources such as files, printers, software, and Internet access.



Communication links:

Communication links in computer networks are the physical or wireless pathways that connect network nodes (like computers, servers, and routers) to transmit data.

Communication devices:

Communication devices in a computer network are physical hardware or software components that transmit and receive data. They form the backbone of network connectivity. Essential devices include modems, routers, switches, hubs, and Network Interface Cards (NICs), each performing distinct roles from directing traffic to translating signals.

Uses of Computer Network

There are multiple uses for a computer network including:

1.Communication:

Through computer networks individuals and organizations can collaborate using communicational channels that may include email, chat, and video conferencing.

2.Resource sharing:

These bags are a boon to users since they provide a way to share the printer, scanner, and files, which will help to improve work activities and reduce costs.

3.Collaboration:

Networks function to make collaboration gin and tonic by offering the opportunities to work jointly on something, share thoughts, and critique in the biggest way.

4.E-commerce:

Online sales and payments processing are empowered with the computer networks, that enable businesses to sell products online and execute secure payments.

5.Education:

From their use in the educational setting they are employed to provide a basis for distance learning, access to resources of higher education and give opportunity for collaboration among students and teachers.

6.Entertainment:

Networks are applied to matters of entertainment like online gaming, online film and music streaming, and social networking.

Advantages of Computer Networks

Here are the advantages uses of a computer network: Here are the advantages uses of a computer network:

- Improved communication and collaboration.
- Resource sharing may play the role of cost-cutting.
- Data quality management and data security.
- Higher automation benefits and remote access possibilities.
- Enhanced productivity and efficiency

Disadvantages of Computer Networks

Here are the disadvantages uses of a computer network: Here are the disadvantages uses of a computer network:

- Data hacking risks and chances of the data leakage.
- The problem of technicalities and operation troubleshooting.
- The likelihood of the exposure of viruses and malware increased.
- There might be a decrease in privacy because of this.

Data Communication:

Data Communication is the process of transferring data between two or more devices through a communication medium (wired or wireless). It allows computers and other devices to exchange information accurately and efficiently.

Example: Sending an email, browsing the internet, or transferring files between two computers.

Types of Data Communication

Based on the direction of data flow, data communication is classified into three types.

1. Simplex Communication

Data flows in one direction only. The receiver cannot send data back.

Examples:

Television Broadcasts, Radio

Advantages:

Simple and inexpensive

Fast transmission

Disadvantages:

No feedback

One-way communication only

2. Half-Duplex Communication

Data can flow in both directions, but only one direction at a time.

Devices take turns transmitting.

Examples

Walkie-Talkie, CB Radio

Advantages

Two-way communication, Less expensive than full duplex

Disadvantages

Delay occurs because devices wait for their turn.

3. Full-Duplex Communication

Data flows in both directions simultaneously. Both devices can send and receive data at the same time.

Examples:

Telephone, Video Calls, Mobile Communication

Advantages

Faster communication

No waiting time

Better performance

Disadvantages

Higher cost

Network Devices or Network Hardware

Network Devices are the physical appliances required for communication and interaction between computers on a computer network.

Enable communication by transmitting and receiving data between devices.

Allow devices to connect to networks efficiently and securely.

Improve network performance by reducing congestion and managing traffic.

Extend network coverage and solve signal loss or attenuation problems.

1.Network Cables

Network cables are the transmission media to transfer data from one device to another. A commonly used network cable is category 5 cable with RJ 45 connector, as shown in the image below:



2.Routers

A router is a connecting device that transfers data packets between different computer networks. Typically, they are used to connect a PC or an organizations LAN to a broadband internet connection. They contain RJ-45 ports so that computers and other devices can connect with them using network cables.



3.Repeaters, Hubs, and Switches

Repeaters, hubs and switches connect network devices together so that they can function as a single segment.

A **repeater** receives a signal and regenerates it before re-transmitting so that it can travel longer distances.

A hub is a multiport repeater having several input/output ports, so that input at any port is available at every other port.

A **switch** receives data from a port, uses packet switching to resolve the destination device and then forwards the data to the particular destination, rather than broadcasting it as a hub.



REPEATER HUB SWITCH

4.Bridges

A **bridge** connects two separate Ethernet network segments. It forwards packets from the source network to the destined network.



5.Gateways

A **gateway** connects entirely different networks that work upon different protocols. It is the entry and the exit point of a network and controls access to other networks.



6.Network Interface Cards

NIC is a component of the computer to connect it to a network. Network cards are of two types: Internal network cards and external network cards.



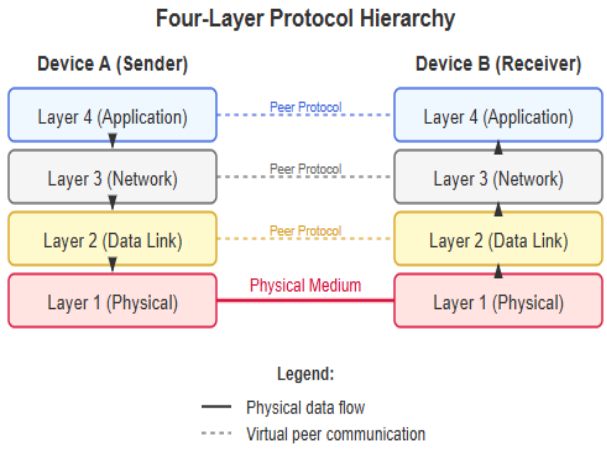
Network software:

Network software is the set of programs, protocols, and services that allow computers and network devices to communicate with each other.

It provides communication rules between different systems.

1. Protocol Hierarchies

Most networks are organized as a stack of layers, one on top of another. The number of layers and their names vary from network to network. Each layer has a specified function and adheres to specified protocols, forming a protocol stack.



How Protocol Hierarchies Work

When Device A wants to send a message to Device B, the data passes through each layer from top to bottom. Each layer performs specific functions on the data and passes it to the layer below. At the physical layer, the bit stream travels through the physical medium to the receiving device.

At Device B, each layer performs its functions according to the protocol shared with its peer layer on Device A, then passes the processed data to the layer above. This continues until the original message reaches the application layer of Device B.

2.Design Issues in Network Layer

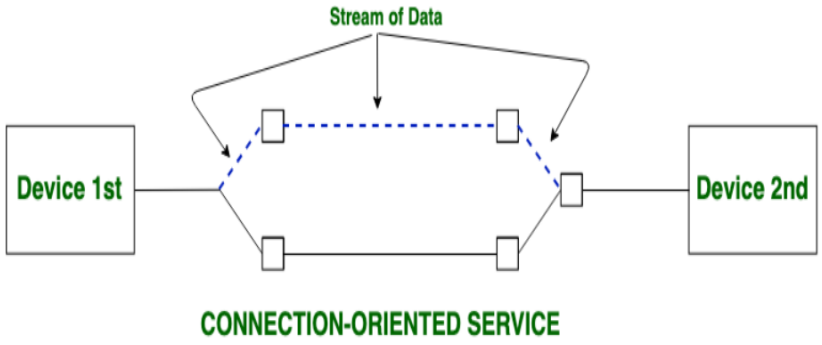
The Network layer is majorly focused on getting packets from the source to the destination, routing error handling, and congestion control. Before learning about design issues in the network layer, let's learn about its various functions.

- **Addressing:** Maintains the address at the frame header of both source and destination and performs addressing to detect various devices in the network.
- **Packeting:** This is performed by Internet Protocol. The network layer converts the packets from its upper layer.
- **Routing:** It is the most important functionality. The network layer chooses the most relevant and best path for the data transmission from source to destination.
- **Inter-networking:** It works to deliver a logical connection across multiple devices.
-

3.Connection-less vs Connection-Oriented

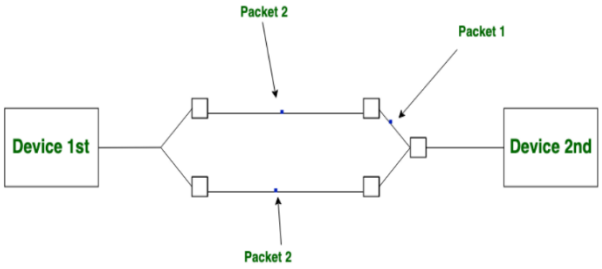
Both Connection-less and Connection-Oriented are used for the connection establishment between two or more devices. These types of services are provided by the Network Layer.

Connection-oriented service: In connection-Oriented service we have to establish a connection between sender and receiver before communication. Handshske method is used to establish a connection between sender and receiver. Connection-Oriented service include both connection establishment as well as connection termination phase. Real life example of this service is telephone service, for conversation we have to first establish a connection**.**



CONNECTION-ORIENTED SERVICE

Connection-less service: In Connection-Less service no need of connection establishment and connection termination. This Service does not give a guarantee of reliability. In this service, Packets may follow the different path to reach their destination. Real life examples of this service is postal system, Online gaming, real-time video and audio streaming etc.



CONNECTIONIESS SERVICE

5. Service primitives:

Service primitives are basic commands or operations that network layers or applications use to communicate. They manage how connections are made, how data moves, and how sessions end. They act like a shared vocabulary between different parts of a network system.

Types of Primitive Actions

- **Request:** A user or layer asks the service to do a task.
- **Indication:** The service tells a target side that an event happened.
- **Response:** The target side replies to the event.
- **Confirm:** The original sender gets proof that the task finished.

6. Relationship Between Services and Protocols

In a computer network, a **service** is a vertical set of capabilities or functions that a layer provides to the layer directly above it, whereas a **protocol** is a horizontal set of rules and formats used by peer entities within the same layer to communicate. Protocols are the hidden mechanisms used to implement and deliver services

Reference Model:

A reference model provides a globally acceptable framework for standardization. Because network users and their devices are spread across widely different locations and may run on entirely different architectures, some kind of standardized reference model is needed to let these heterogeneous devices communicate regardless of their underlying design.

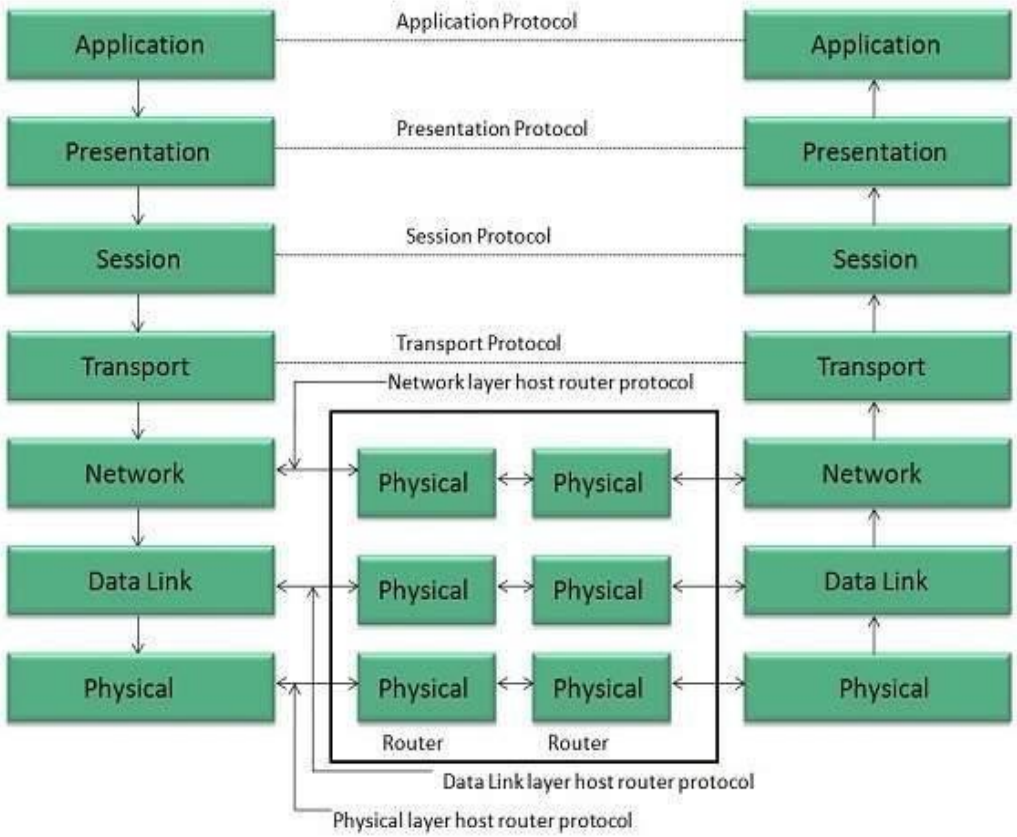
Two reference models are widely used — the OSI model and the TCP/IP model. The OSI model is largely theoretical, whereas TCP/IP is the model actually put into practice.

ISO- OSI Reference Model:

OSI stands for Open System Interface. It was developed by the International Organization for Standardization (ISO), which is why it is also called the ISO-OSI model.

As the diagram below shows, the OSI model is built from seven layers, each with its own specific job, and each providing services to the layer above it.

- Physical layer
- Data link layer
- Network layer
- Transport layer
- Session layer
- Presentation layer
- Application layer



Physical Layer:

- The Physical Layer sits at the very bottom of the OSI Model.
- It handles the actual sending and receiving of raw, unstructured bits across the network.
- It performs encoding and decoding — converting bits into the appropriate signal form.
- It turns digital or analog data into electrical or optical signals for transmission.
- Data moves through this layer purely as bits.
- It also determines whether a link operates as simplex, half-duplex, or full-duplex.

Data Link Layer:

- The Data Link Layer coordinates the information that will be sent out over the physical layer.
- Its main job is guaranteeing error-free transfer of data from one node to the next across the physical layer.
- Data at this layer is organized into frames.
- It manages the sequential transmission and reception of these frames.
- It sends and expects acknowledgements for frames that are transmitted and received.
- It adds error-detection and error-correction information to outgoing data.

Network Layer:

- The Network Layer handles delivering data between hosts that sit on different networks.
- It also manages packet routing — picking the shortest available path from among the possible routes.
- (This involves the IP address — source plus destination — along with port addresses and fragmentation.)
- It effectively acts as the network's controller.
- It oversees traffic within the subnet.
- It determines which route the data should follow.
- Data at this layer takes the form of packets.
- It carries out routing — its protocols determine the best path from source to destination.

Transport Layer:

- It decides whether data should travel over a single path or multiple parallel paths.
- It provides end-to-end data delivery by breaking data into segments (with a port address attached to each fragment).

- It carries out multiplexing and splitting of the data (combining port addresses with fragments).
- It divides data into smaller units so the network layer can handle it more efficiently.
- It also supplies flow control, error control, sequence numbering, and message acknowledgement to ensure reliable delivery.
- Data at this layer is organized into datagrams — self-contained packets carrying their own source and destination information for correct routing.

Session Layer:

The Session layer carries out the following functions:

- Synchronization: this layer lets a process insert checkpoints into the data stream, functioning as synchronization points that help reduce errors.
- These checkpoints make it easier to pinpoint an error so the data can be properly re-synchronized.
- Dialog Controller: the session layer lets two systems begin communicating in either half-duplex or full-duplex mode.
- It is responsible for establishing, managing, and terminating sessions.

Presentation Layer:

The Presentation layer carries out the following functions:

- It is also known as the Translation layer.
- Its translation duties include encrypting and decrypting data (converting between plaintext and cipher text) as well as compressing it.

The functions of the presentation layer are:

Translation Encryption/ Decryption: Encryption converts data into another form or code; the result is called cipher text, and reversing the process yields plain text again.

Compression: Cuts down the number of bits that need to travel across the network.

Application Layer:

The Application layer carries out the following functions:

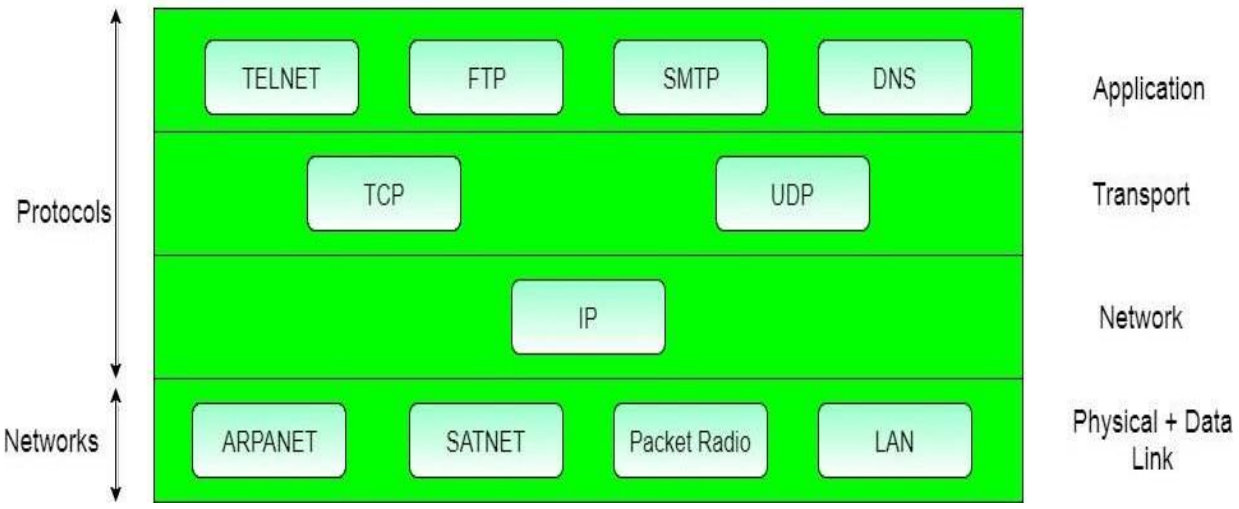
- It sits at the very top of the OSI Reference Model's stack of layers.

- It gives users access to network resources.
- Applications communicate directly with the network's data through tools such as web browsers.

- It offers various services, including manipulating information in different ways, retransmitting files, and distributing results across the network.
- It also carries out the authentication needed to perform transactions.

TCP/IP Model:

- The TCP/IP model is the practical model actually used on the Internet. TCP/IP stands for Transmission Control Protocol and Internet Protocol.
- It merges the OSI model's Physical and Data Link layers into a single Host-to-Network layer. The diagram below shows its layers:
    - Link layer or Host to network.
    - Internet layer
    - Transport layer
    - Application layer



Application Layer:

- This layer mirrors its OSI counterpart and performs the following functions:

- It offers services such as manipulating information in various ways, retransmitting files, and distributing results.
- Functions like login and password verification are also handled here.
- Protocols used: TELNET, FTP, SMTP, HTTP, and SNMP all operate at this layer.

Protocols in application layer:

HTTP: Short for Hypertext Transfer Protocol, it lets us access data across the World Wide Web, carrying plain text, audio, and video content.

SMTP: Short for Simple Mail Transfer Protocol, it is the TCP/IP protocol underlying e-mail and is used to deliver messages to another e-mail address.

DNS: Short for Domain Name System, it uses an IP address to uniquely identify a host's connection to the Internet.

TELNET: Short for Terminal Network, it sets up a connection between a local computer and a remote one.

FTP: Short for File Transfer Protocol, it is a standard Internet protocol used to move files between computers.

Transport Layer:

- It performs the same role as the transport layer in the OSI model. Key points include:
- It relies on TCP and UDP for end-to-end transmission.
- TCP is connection-oriented and also manages flow control.
- UDP, on the other hand, is connectionless.
- It segments and splits data as needed.
- It decides whether data travels over one path or multiple paths.
- It attaches header information to ensure reliable, error-free, end-to-end delivery.
- Protocols used: TCP/IP and UDP operate at this layer.

Internet Layer:

This layer lets a host inject packets into the network and have them travel independently toward their destination — though they may not necessarily arrive in the order they were sent (which relates to packet delivery, routing, and congestion control).

- The Internet layer is the second layer of the TCP/IP model.

- It's also known as the network layer.
- Its main job is delivering packets from any network so they reach their destination regardless of the independent route each one takes.

Protocols used:

IP (Internet protocol)

Handles path selection, routing, and addressing

- ICMP (Internet Control Message Protocol)
- ICMP Test: used to check whether a destination is reachable.
- ICMP Reply: used to check whether the destination device is responding.

Host-to-Network Layer:

- This is the lowest layer of the TCP/IP model.
- It combines the OSI reference model's Physical layer and Data Link layer into one.
- It handles the physical transmission of data.
- It is mainly responsible for physically moving data between two devices on the network.
- It defines the protocols needed to connect a host.

Ethernet: Ethernet refers to a family of networking technologies widely used across local area networks, metropolitan area networks, and wide area networks.

DSL: Short for Digital Subscriber Line, it gives users a high-speed bandwidth connection over a network.

SONET: Synchronous Optical Network is a communication protocol used to move large volumes of data over long distances via optical fiber.

Difference b/w OSI Reference model VS TCP/IP :

OSI (Open System Interconnection)	**TCP/IP (Transmission Control Protocol / Internet Protocol)**
A generic, protocol-independent standard that acts as a communication gateway between the network and the end user.	Built around standard protocols on which the Internet itself developed — a communication protocol that lets hosts connect across a network.

The transport layer in the OSI model guarantees packet delivery.	The transport layer in TCP/IP does not guarantee delivery, yet the TCP/IP model is still considered more reliable overall.
Follows a vertical approach.	Follows a horizontal approach.
Includes separate Presentation and Session layers.	Has no distinct Presentation or Session layer.
The Transport Layer is connection-oriented.	The Transport Layer supports both connection-oriented and connectionless modes.
The Network Layer supports both connection-oriented and connectionless modes.	The Network Layer is connectionless.

OSI is a reference model that networks are built around and is mainly used as a guidance tool.	TCP/IP is, in a sense, an implementation of the OSI model.
The OSI model's Network layer offers both connection-oriented and connectionless service.	The Network layer in TCP/IP only offers connectionless service.
OSI has trouble fitting real protocols cleanly into its model.	TCP/IP isn't tied to any single protocol in the same way.
Protocols are hidden within the OSI model and can be swapped out easily as technology evolves.	Replacing a protocol in TCP/IP is not straightforward.
OSI clearly defines services, interfaces, and protocols as separate concepts and is protocol-independent.	In TCP/IP, services, interfaces, and protocols are not cleanly separated, and it is protocol-dependent.
Has 7 layers	Has 4 layers

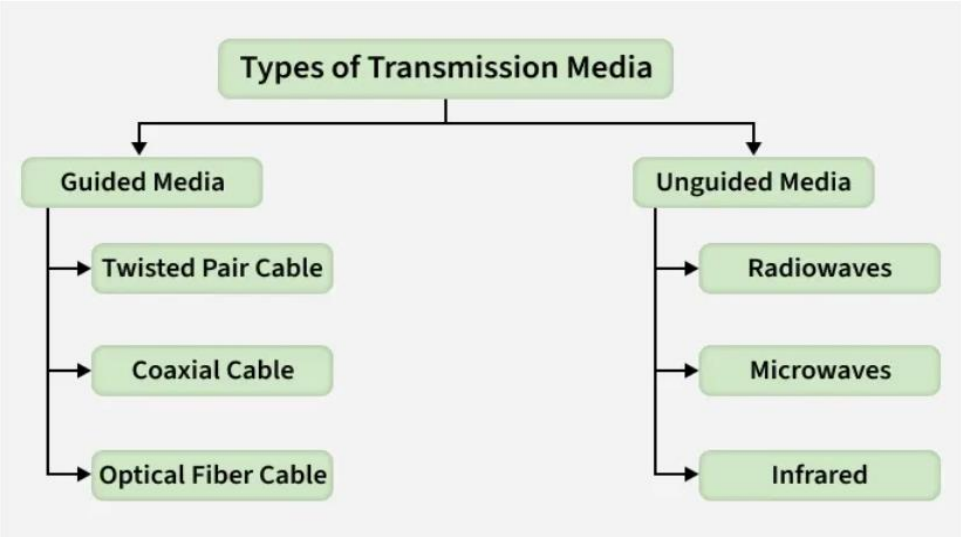
Transmission media:

Transmission media refers to the physical or wireless communication channel used to carry data signals from one device to another within a computer network. It forms the fundamental pathway through which information is transmitted, ensuring connectivity between networked devices.

The selection of a transmission medium depends on factors such as:

- Transmission distance
- Data transfer speed (bandwidth)
- Susceptibility to interference and noise
- Cost and installation requirements

Based on the nature of the transmission path, transmission media are broadly classified into two main types:



Guided Media or bounded :

Guided Media also known as wired or bounded transmission media, refers to transmission media in which data signals are transmitted through a physical path using cables. The signal is confined and guided along a fixed route, providing controlled communication between network devices.

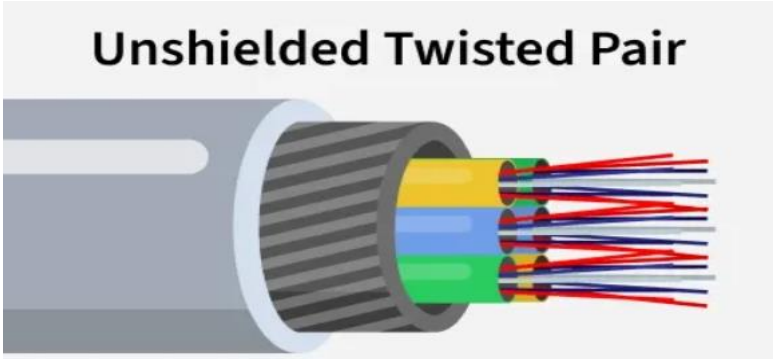
- Uses physical cables to transmit data signals.
- Provides dedicated and well-defined transmission paths.
- Major types of guided media included Twisted Pair Cables, Coaxial Cables and Optical Fiber Cables.
- Offers higher data transmission rates compared to most wireless media.
- Provides better security due to physical connectivity and limited signal leakage.
- Suitable for short to long-distance communication, depending on the cable type used.

1. Twisted Pair Cable

A twisted pair cable consists of two individually insulated copper conductors twisted together in a helical pattern to minimize electromagnetic interference (EMI) and crosstalk between adjacent pairs. Multiple twisted pairs are typically enclosed within a protective outer sheath, making this cable one of the most commonly used and cost-effective guided transmission media.

- Twisting of conductors helps reduce interference and signal degradation.
- Multiple pairs can be bundled together within a single cable.
- Widely used due to low cost, flexibility, and ease of installation.
- Commonly used in telephone systems and local area networks (LANs).

(a) **Unshielded Twisted Pair (UTP):** UTP cable consists of twisted copper wire pairs without any additional metallic shielding. The twisting provides basic protection against interference, making UTP suitable for environments with moderate noise levels.



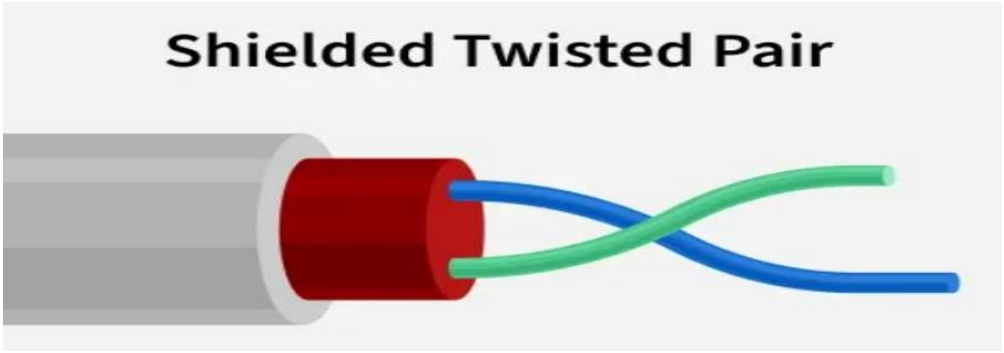
Advantages:

- Least expensive among guided transmission media.
- Easy to install and maintain due to its lightweight and flexible nature.
- Supports high data transmission rates over short distances.
- Widely used in Ethernet-based LANs and telephone networks.

Disadvantages:

- Lower performance and capacity compared to Shielded Twisted Pair (STP).
- Limited transmission distance due to signal attenuation.
- More susceptible to noise and electromagnetic interference.

(b) Shielded Twisted Pair (STP): Shielded Twisted Pair (STP) cable consists of twisted pairs of insulated copper conductors enclosed within an additional metallic shielding layer, such as a foil or braided shield. This shielding provides enhanced protection against external electromagnetic interference (EMI) and reduces crosstalk, resulting in more stable and reliable data transmission.



Advantages

- Provides better performance at higher data transmission rates compared to UTP.
- Significantly reduces electromagnetic interference and crosstalk.
- Ensures more reliable and stable communication, especially in noisy environments.
- Suitable for high-speed Ethernet networks and voice and data transmission systems.

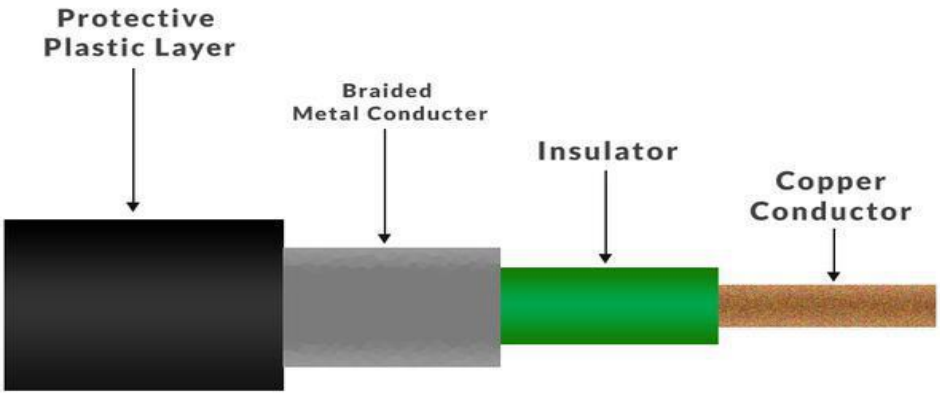
Disadvantages

- Higher cost compared to Unshielded Twisted Pair (UTP).
- More complex installation and termination due to shielding requirements.
- Bulkier and less flexible, making cable management more difficult.

2. Coaxial Cable

Coaxial Cable consists of a central copper conductor surrounded by a dielectric insulating layer, a metallic shielding layer, and an outer protective jacket. This structure allows signals to be transmitted with better protection against noise and interference compared to twisted pair cables.

- Supports data transmission in both baseband mode (single channel) and broadband mode (multiple channels).
- Provides higher bandwidth than twisted pair cables.
- Strong shielding makes it more resistant to noise, crosstalk, and electromagnetic interference (EMI).
- Commonly used in cable television (CATV), broadband networks, and analog television systems.
- More durable and reliable due to its layered construction.
- Easier to install and maintain compared to optical fiber cables.



Advantages:

- Supports higher bandwidth than twisted pair cables.
- Offers better signal quality and reliability.
- Allows multiple channels using frequency division multiplexing (FDM).
- Less affected by external electromagnetic interference.

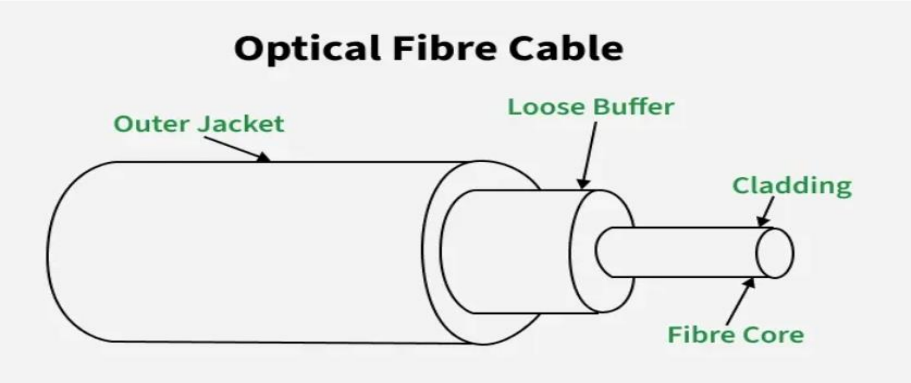
Disadvantages:

- More expensive than twisted pair cables.
- Requires proper grounding to ensure safety and minimize interference.
- Bulkier and less flexible due to multiple layers.
- More vulnerable to security breaches, as the cable can be physically tapped.

3. Optical Fiber Cable

Optical Fiber Cable is a guided transmission medium that transmits data in the form of light signals through a glass or plastic core using the principle of total internal reflection. The core is surrounded by a cladding layer with a lower refractive index, which confines the light within the core and enables high-speed, long-distance data transmission.

- Supports very large data volumes at extremely high speeds.
- Can operate in unidirectional or bidirectional communication modes.
- WDM (Wavelength Division Multiplexer) allows multiple light signals to be transmitted simultaneously over a single fiber.
- Widely used where high bandwidth, long distance, and low signal loss are required.



Advantages:

- Provides very high bandwidth and data-carrying capacity.
- Lightweight and compact, making installation easier over long distances.
- Exhibits low signal attenuation, suitable for long-distance communication.
- Immune to electromagnetic interference (EMI) and electrical noise.
- Resistant to corrosion and harsh environmental conditions.

Disadvantages:

- Complex installation and maintenance due to precise splicing and handling requirements.
- High initial installation and equipment cost.
- Fragile compared to copper cables and requires careful handling.

Applications:

- **Medical:** Used in endoscopy, imaging systems, and laser-based surgical procedures.
- **Defense and Aerospace:** Supports secure, high-speed, and interference-free communication.
- **Communication:** Widely used in internet backbone networks, undersea cables, and telecommunication systems.
- **Industrial and Automotive:** Used for sensing, lighting, data communication, and safety systems.

Unguided Media or Unbounded

Unguided media, also known as wireless or Unbounded transmission media , uses electromagnetic waves to transmit data without any physical medium. Signals propagate through free space such as air or vacuum. The main types of unguided media are radio waves, microwaves, and infrared waves.

Features:

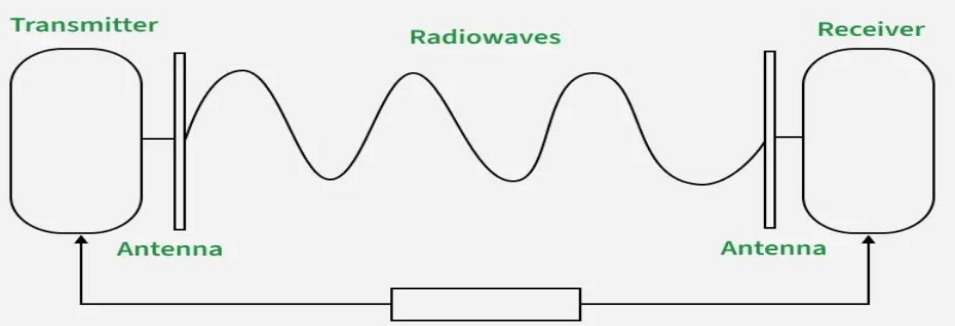
- Signals propagate through air or free space
- Less secure due to broadcast nature
- Suitable for long-distance communication

1. Radio Waves

Radio waves are a type of electromagnetic wave that can easily be generated and can propagate through buildings and other obstacles. They do not require line-of-sight between transmitting and receiving antennas, making them highly suitable for broadcast communication and wireless data transmission.

Frequency Range:

3 kHz – 300 GHz



Applications:

AM and FM radio broadcasting, television transmission, cordless phones, and wireless communication.

Types of Radio Waves:

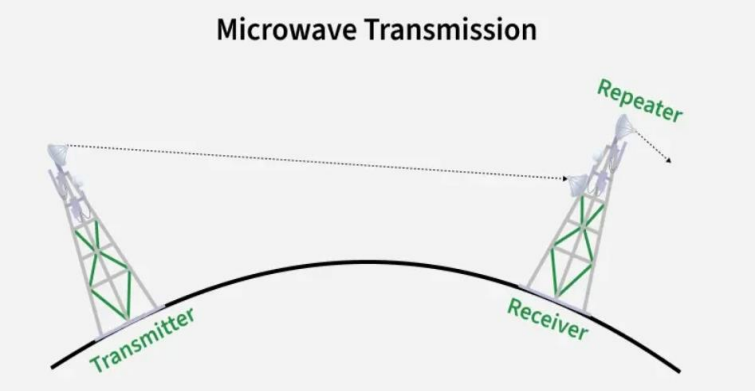
- **Shortwave:** AM radio broadcasting
- **VHF (Very High Frequency):** FM radio and television
- **UHF (Ultra High Frequency):** Television and mobile communication.

Components:

- **Transmitter:** Generates and modulates the signal
- **Receiver:** Receives and demodulates the signalRadiowaves

2. Microwaves

Micro waves are a form of unguided transmission media that use line-of-sight communication, where the transmitting and receiving antennas must be properly aligned. The transmission range depends on the height of the antennas. Microwaves operate in the frequency range of 1 GHz to 300 GHz and are widely used in mobile communication, satellite links, and television distribution.



Advantages:

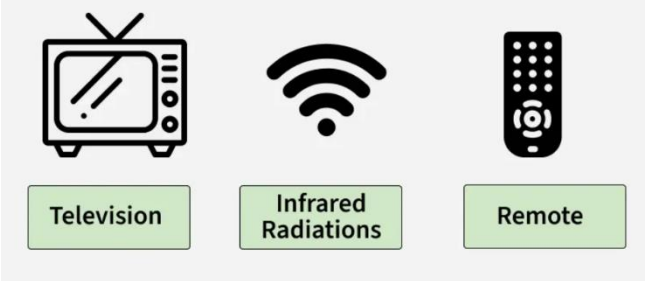
- Cost-effective compared to laying physical cables
- No need for land acquisition
- Suitable for communication over difficult terrains and oceans
- Supports high data transmission rates

Disadvantages:

- Less secure without proper encryption
- Susceptible to weather conditions such as rain and fog
- Affected by obstacles due to line-of-sight requirement
- High cost of antenna design, installation, and maintenance

3. Infrared

Infrared waves are used for short-range wireless communication and operate in the frequency range of 300 GHz to 400 THz. They cannot penetrate solid obstacles, which limits their range but helps minimize interference between nearby systems. Infrared waves are commonly used in TV remote controls, wireless keyboards, mice, and printers.



Difference Between Radio Waves, Micro Waves, and Infrared Waves

Radio Waves	Microwaves	Infrared Waves
Omnidirectional in nature	Highly directional	Directional, requires line of sight (LOS)
Can penetrate buildings and obstacles easily due to low frequency	Poor penetration; line of sight required	Cannot penetrate obstacles
Frequency range: 3 kHz – 300 GHz	Frequency range: 1 GHz – 300 GHz	Frequency range: 300 GHz – 400 THz
Low security	Medium security (encryption possible)	Relatively high security but limited range
Moderate attenuation	Variable attenuation, depends on weather conditions	High attenuation
Some frequency bands require government licensing	Most frequency bands are licensed by the government	Generally unlicensed
Low to moderate cost	High cost	Low cost
Supports long-distance communication	Supports long-distance communication	Suitable only for very short-distance communication

ARPANET stands for **Advanced Research Projects Agency NET**. ARPANET was the first network that consisted of distributed control. It was the first to implement TCP/IP protocols. It was basically the beginning of the Internet with the use of these technologies. It was designed with a basic idea in mind that was to communicate with scientific users among an institute or university.

History of ARPANET

ARPANET was introduced in the year 1969 by the Advanced Research Projects Agency (ARPA) of the US Department of Defense. It was established using a bunch of PCs at various colleges and sharing of information and messages was done. It was for playing as long separation diversions and individuals were asked to share their perspectives. In the year 1980, ARPANET was handed over to a different military network, Defense Data Network.

Characteristics of ARPANET

1. It is basically a type of WAN.
2. It used the concept of a packet-switching network.
3. It used Interface Message Processors(IMPs) for sub-netting.
4. ARPANETs software was split into two parts- a host and a subnet.

Advantages of ARPANET

- ARPANET was designed to serve even in a Nuclear Attack.
- It was used for collaborations through E-mails.
- It created an advancement in the transfer of important files and data for defense.

Limitations of ARPANET

- Increased number of LAN connections resulted in difficulty handling.
- It was unable to cope-up with advancement in technology.

What is Internet

The Internet is a global system that connects computers and devices worldwide, enabling communication, information sharing, and access to digital services.

- Connects people and devices globally
- Enables communication, learning, and business
- Supports digital services and cloud platforms
- Operates using standard communication protocols

- Forms the backbone of modern digital life

History of the Internet

The Internet originated in the 1960s with the development of ARPANET (Advanced Research Projects Agency Network). It was designed to allow multiple computers to communicate within a single network an achievement that was revolutionary at the time.

ARPANET used packet switching, a method that breaks data into packets for efficient transmission. In October 1969, the first message was successfully sent between two computers using ARPANET. From that moment, internet technology continued to evolve rapidly.

How is the Internet Set Up?

The Internet relies on physical infrastructure such as:

- Optical fiber cables
- Copper wires
- Networking systems like LAN, WAN, and MAN

Mobile technologies like 2G, 3G, 4G, and Wi-Fi also depend on these physical cable networks to function.

The organization ICANN (Internet Corporation for Assigned Names and Numbers), based in the USA, manages essential Internet components such as IP addresses, the Domain Name System (DNS), and global protocols to ensure smooth connectivity.

How Does the Internet Work?

The Internet operates using a client–server model:

- **Client:** A device such as a laptop or smartphone connected to the Internet
- **Server:** Powerful computers that store websites and data

Step-by-step working:

1. The client sends a request through a browser
2. The server identifies the website using its IP address
3. DNS converts the domain name into an IP address
4. The request is routed to the correct server
5. The server processes the request and returns the webpage

Example:

Just as a name can help locate a person's Aadhaar number, a domain name helps DNS find the correct IP address.